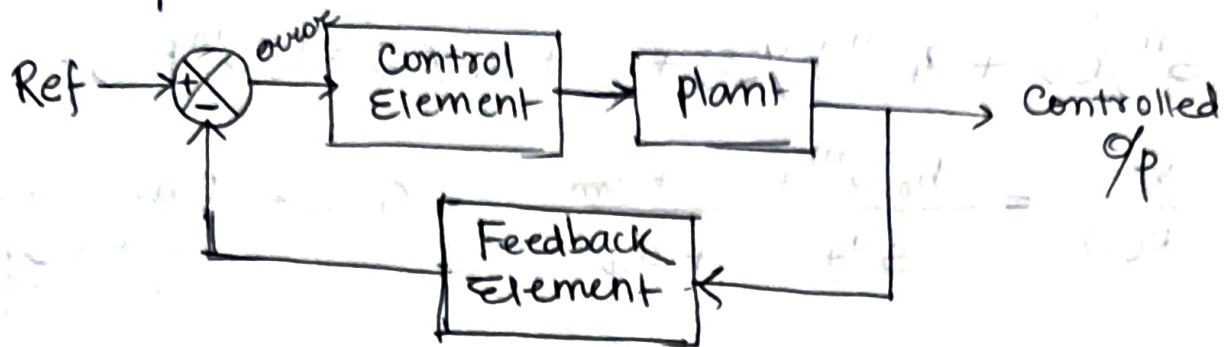


Control System:-

• Closed loop control:-



• Manually controlled closed-loop sys:



• System Models:-

1) Dynamic Eqn of any physical system is given by -

$$a_0 \frac{d^n y}{dt^n} + a_1 \frac{d^{n-1} y}{dt^{n-1}} + a_2 \frac{d^{n-2} y}{dt^{n-2}} + \dots + a_{n-1} \frac{dy}{dt} + a_n y =$$

$$b_0 \frac{d^m r}{dt^m} + \dots + b_{m-1} \frac{dr}{dt} + b_m r$$

Where,

$y = y(t)$, o/p Variable

$r = r(t)$, i/p

$a_i, b_j \rightarrow f^n$ of sys parameters.

$$\boxed{n \geq m}$$

System must be causal.

→ The system o/p's are determined by past i/p's and not by future i/p's is called a Causal system.

e.g:- $n=0, m=1$

$$a_0 y = b_0 \frac{dr}{dt} + b_1 r, \text{ let } a_0 = b_0 = b_1 = 1 \rightarrow \text{future i/p not a causal}$$

$$y = \frac{dr}{dt} + r \Rightarrow y(t) = r(t) + \frac{r(t+t_2) - r(t-t_2)}{t}$$

2) Taking d.T with zero initial condⁿ:-

$$\Rightarrow s^n y(s) + a_1 s^{n-1} y(s) + \dots + a_n y(s) = b_0 s^m R(s) + \dots + b_m R(s)$$

$$\frac{Y(s)}{R(s)} = \frac{b_0 s^m + \dots + b_m}{s^n + \dots + a_n} = G(s) \leftarrow \text{Transfer function of the sys.}$$

$$G(s) = \frac{N(s)}{A(s)} = \frac{K (s+z_1) (s+z_2) \dots (s+z_m)}{(s+p_1) (s+p_2) \dots (s+p_n)}$$

$\Delta(s) \rightarrow$ characteristic polynomial.

Characteristic Eqⁿ $\Rightarrow \Delta(s) = 0 \rightarrow s = -p_1, -p_2, -p_3, \dots, -p_n$ (poles)
 $N(s) = 0 \rightarrow s = -z_1, -z_2, \dots, -z_m$ (zeros)

* Transfer function of mass Spring damper sys.

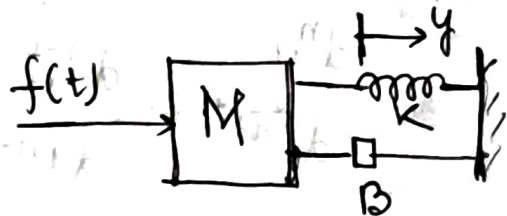
f : force

y : displacement

M : Mass

K : Spring const.

B : damping coefficient.



$$f(t) = Ky + B \frac{dy}{dt} + M \frac{d^2y}{dt^2}$$

Taking the d.T with zero initial condⁿ.

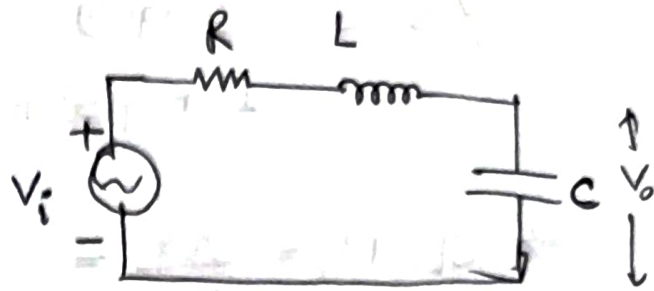
$$Ms^2 y(s) + B s y(s) + K y(s) = F(s)$$

$$\Rightarrow G(s) = \frac{Y(s)}{F(s)} = \frac{1}{Ms^2 + Bs + K}$$

* Transfer function of series RLC ckt:

$$V_i = iR + L \frac{di}{dt} + V_o, \quad i = C \frac{dV_o}{dt}$$

$$\Rightarrow V_i = RC \frac{dV_o}{dt} + LC \frac{d^2V_o}{dt^2} + V_o$$



d.T with zero initial condⁿ

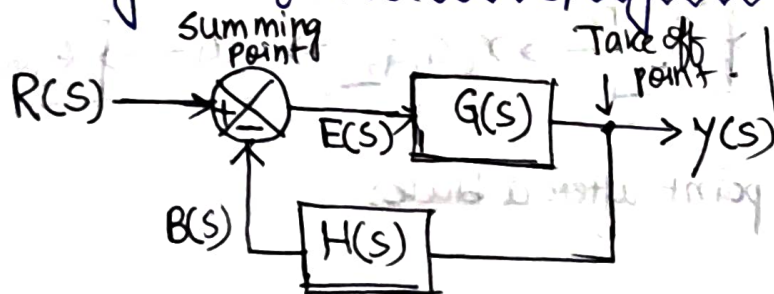
$$\Rightarrow V_i(s) = RC s V_o(s) + LC s^2 V_o(s) + V_o(s)$$

$$\Rightarrow \frac{V_o(s)}{V_i(s)} = \frac{1}{LC s^2 + RC s + 1} = \frac{1/LC}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

* Block Diagram:-

$$Y(s) = G(s) R(s)$$

* Block diagram of a closed loop system:-



$R(s)$: Ref i/p

$Y(s)$: Controlled o/p

$G(s)$: Forward path TF

$H(s)$: Feedback path TF

$E(s)$: Error signal

$B(s)$: Feedback signal.

$$Y(s) = G(s) E(s)$$

$$E(s) = R(s) - B(s)$$

$$= R(s) - H(s) Y(s)$$

$$\therefore Y(s) = G(s) [R(s) - H(s) Y(s)]$$

$$Y(s) = G(s) R(s) - G(s) H(s) Y(s)$$

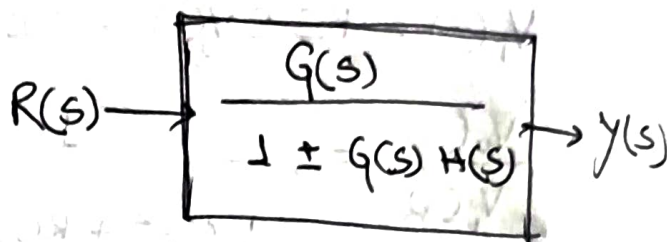
$$\Rightarrow [1 + G(s) H(s)] Y(s) = G(s) R(s)$$

$$Y(s) = \frac{G(s)}{1 + G(s)H(s)} R(s)$$

$$\rightarrow \text{CL. T.f.} = \frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

General,

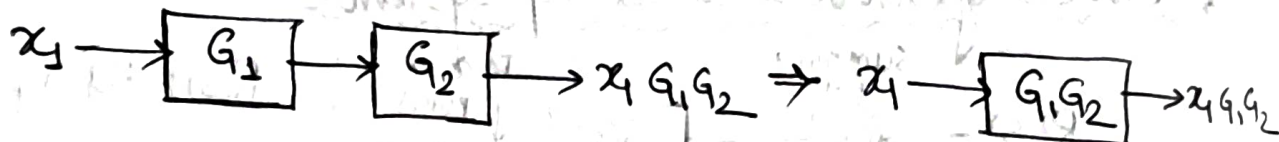
$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 \pm G(s)H(s)}$$



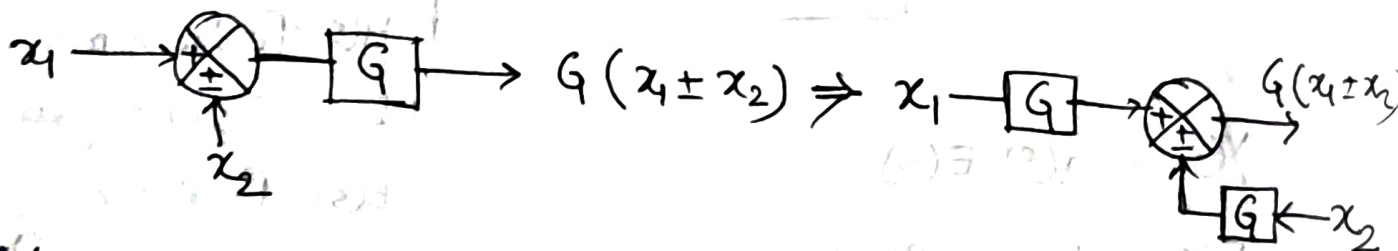
T.F for closed loop system.

* Block diagram reduction rule:

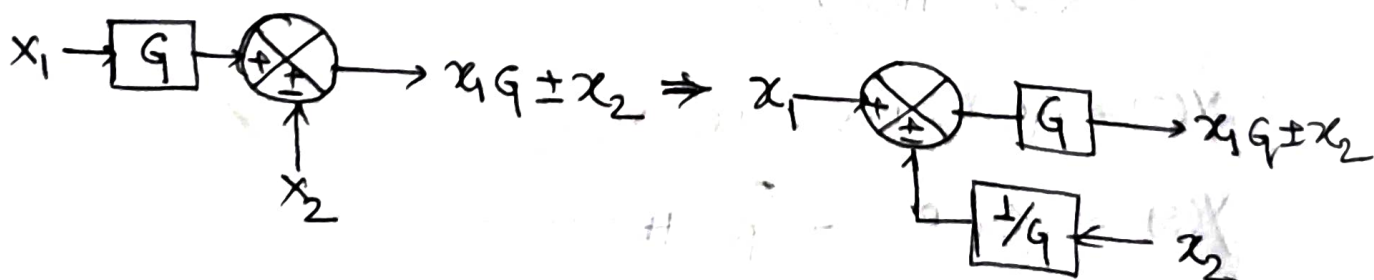
1. Combining blocks in cascade:



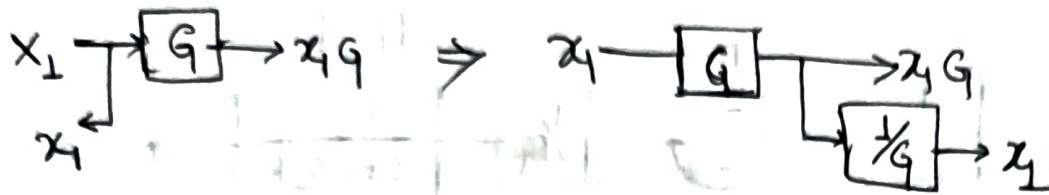
2. Moving a summing point after a block:



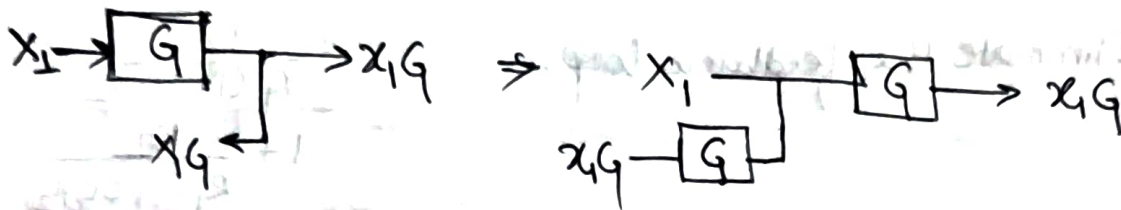
3. Moving a summing point ahead of a block:



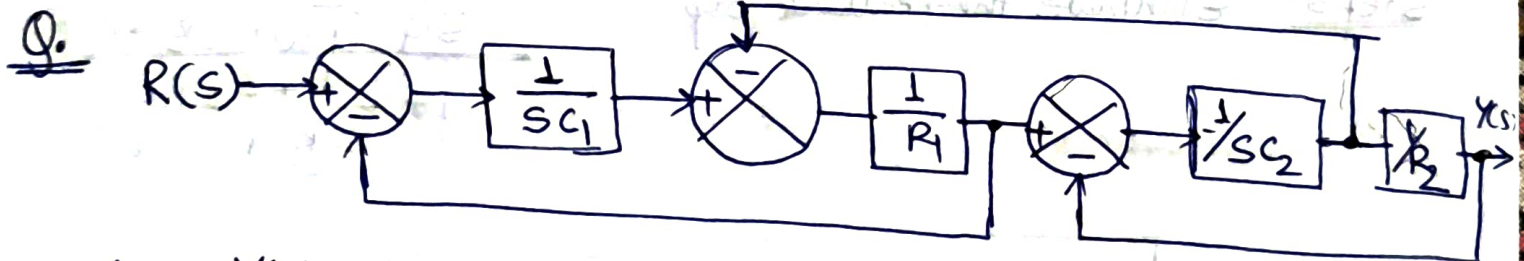
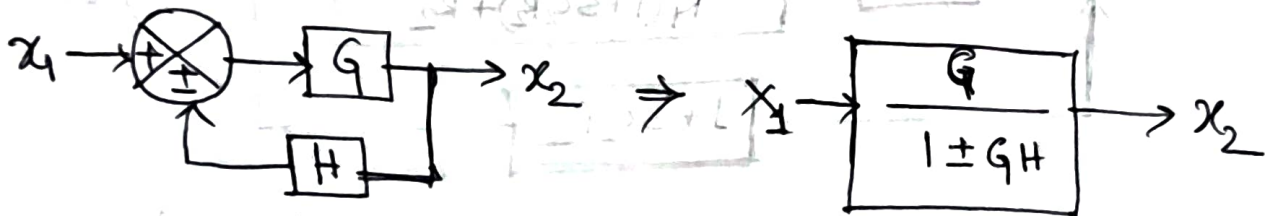
4. Moving a takeoff point after a block:-



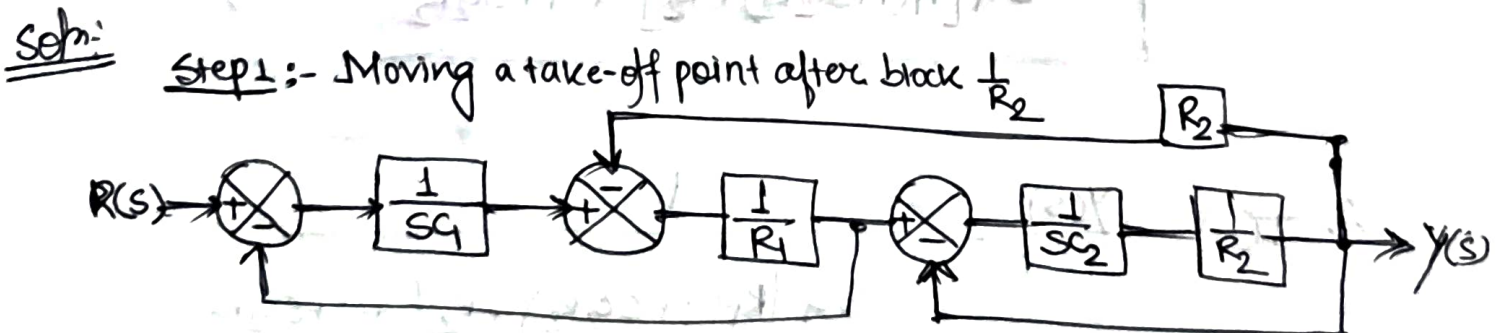
5. Moving a take off point ahead of a block.



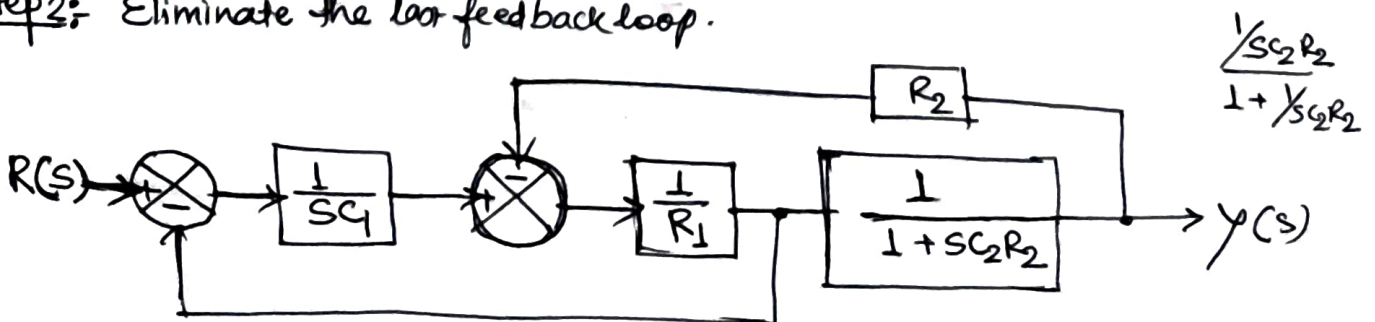
6. Eliminating feedback loop:



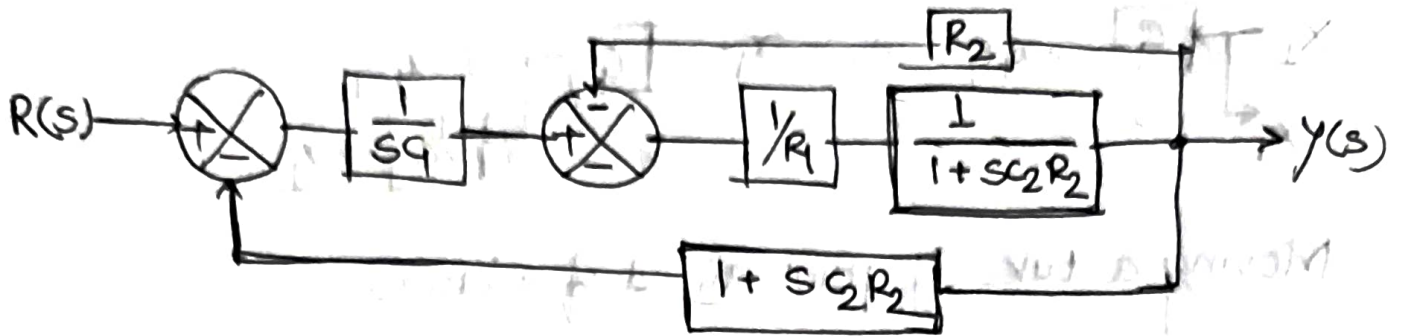
Find $\frac{Y(s)}{R(s)}$?



Step 2:- Eliminate the last feedback loop.

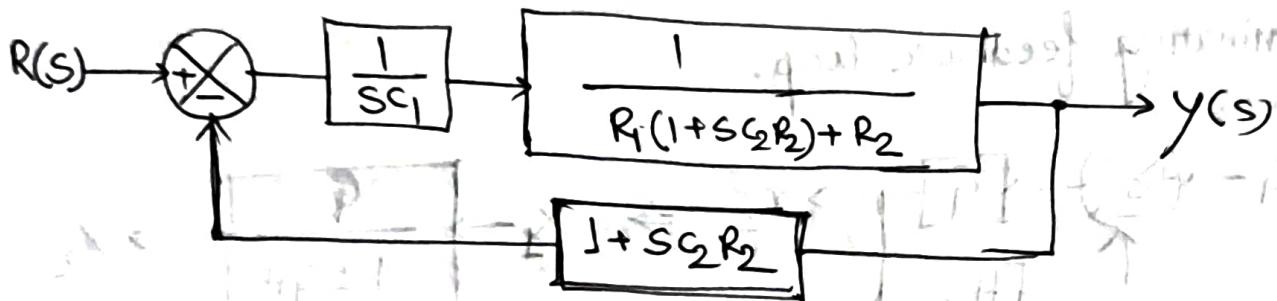


Step 3: Move the take-off point after the block $\frac{1}{1+sG_2R_2}$



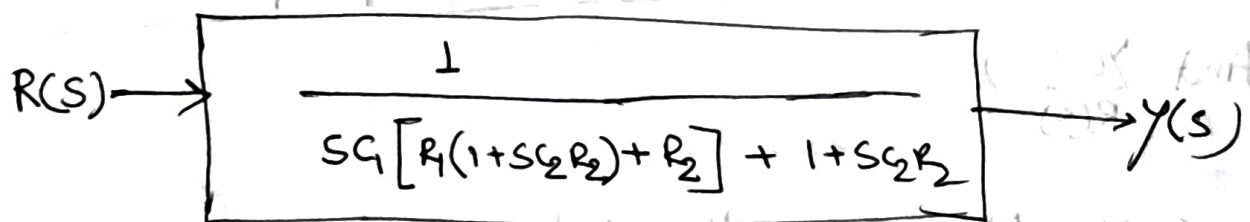
Step 4: Eliminate the feedback loop.

$$\frac{\frac{1}{R_1(1+sG_2R_2)}}{1 + \frac{R_2}{R_1(1+sG_2R_2)}}$$



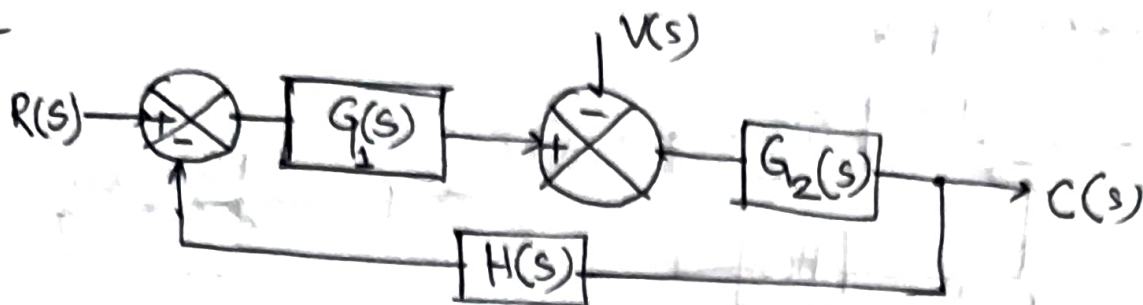
Step 5: Eliminate the feedback loop.

$$\frac{\frac{1}{sG_1} \frac{1}{[R_1(1+sG_2R_2) + R_2]}}{1 + \frac{1+sG_2R_2}{sG_1[R_1(1+sG_2R_2) + R_2]}}$$

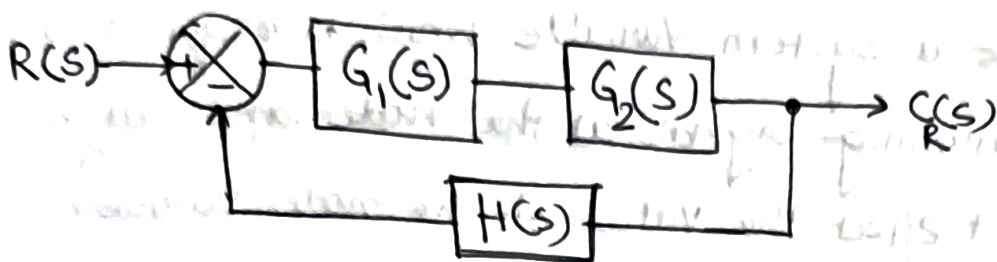


$$\therefore T(s) = \frac{Y(s)}{R(s)} = \frac{1}{sG_1 [R_1(1+sG_2R_2) + R_2] + 1 + sG_2R_2}$$

eg:-

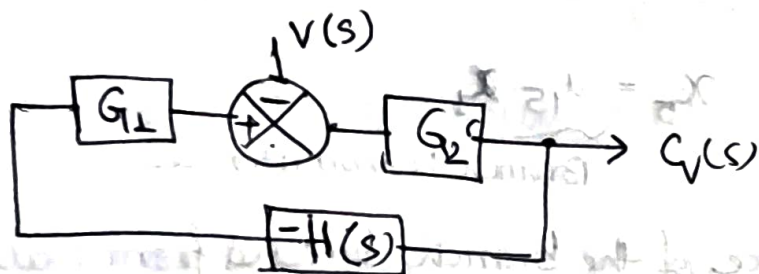


Soln:- When, $V(s) = 0$



$$C_R(s) = \frac{G_1 G_2}{1 + G_1 G_2 H} R(s)$$

When $R(s) = 0$



$$C_V(s) = \frac{G_2 (-V(s))}{1 + G_2 G_1 (-H)}$$

$$C_V(s) = \frac{-G_2 V(s)}{1 + G_1 G_2 H}$$

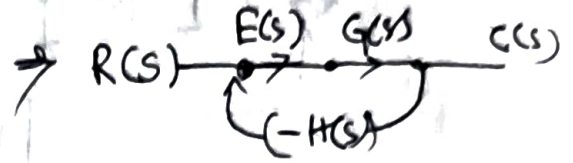
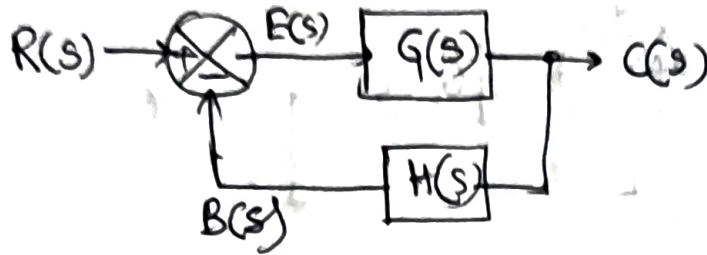
For linear system superposition thm

$$C(s) = C_R(s) + C_V(s)$$

$$= \frac{G_1 G_2 R(s)}{1 + G_1 G_2 H} + \frac{(-G_2 V(s))}{1 + G_1 G_2 H}$$

$$= \frac{G_2}{1 + G_1 G_2 H} [G_1 R(s) - V(s)]$$

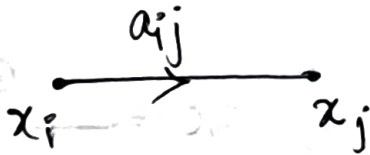
* Signal flow graph:-



1) Node:-

It represents a system variable which is required to the sum of all incoming signals at the node and outgoing signals do not effect the values of the node variable.

2) Branch:- Signal travels along a branch from one node to another in the direct indicated by the branch arrow and in the process gets multiplied by the gains or transmittance of the branch



$$x_5 = a_{45} x_1$$

Branch transmittance

a_{ij} is the transmittance of the branch directed from node x_i to x_j

3) Input node or Source:- It is a node with only outgoing branches.

4) Output node:- It is a node with only incoming branches, however this condⁿ is not always met and additional branch with unity gain may be introduced in order to met this specified condⁿ.

5) Path:- It is the traversal of connected branch in the directⁿ of branch arrow, such that no node is traversed more than once.

6) Forward path:- It is a path from the i/p node to the o/p node.

7) Loop: It is a path which originates and terminates at the same node.

8) Non-touching loop:- Loops are said to be non touching if they do not share any common node.



① & ② non-touching

9) Forward path gain:- It is the product of the branch gain encountered in traversing a forward path.

10) Loop Gain:- It is a product of branch gain encountered in traversing the loop.

* Construction of Signal flow graph:

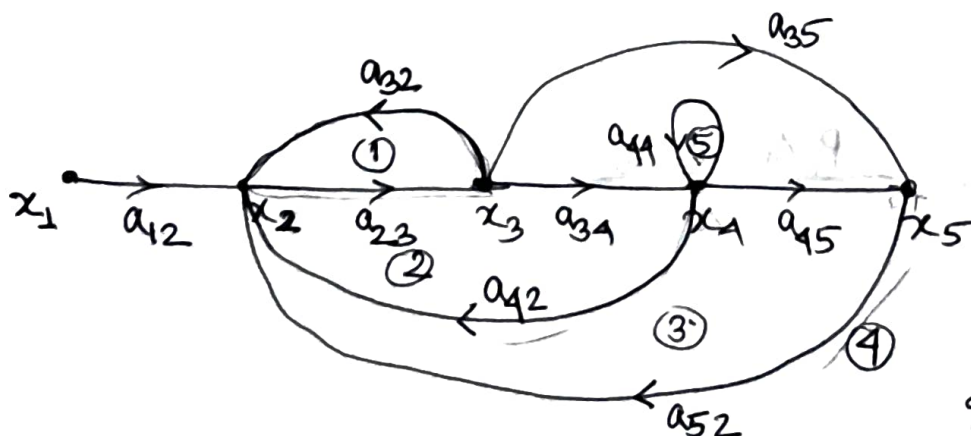
$$x_2 = a_{12} x_1 + a_{32} x_3 + a_{42} x_4 + a_{52} x_5$$

$$x_3 = a_{23} x_2$$

$$x_4 = a_{34} x_3 + a_{44} x_4$$

$$x_5 = a_{35} x_3 + a_{45} x_4$$

Where x_1 is the i/p variable and x_5 is the o/p variable



graph 1

* Mason's Gain Formula

$$T = \frac{1}{\Delta} \sum_k P_k \Delta_k$$

Where,

$K \rightarrow$ No. of Forward path

$P_k \rightarrow$ path gain of k^{th} forward path.

$\Delta_k =$ the value of Δ not touching with the k^{th} forward path

$\Delta =$ determinant of the graph

$$= 1 - (\text{sum of loop gains}) +$$

(sum of gain products of two non-touching loop)

+ (sum of ---)

Q. Find $\frac{x_5}{x_1}$? from the signal flow graph.

Soln Here $K=2$

$$P_1 = a_{12} a_{23} a_{34} a_{45}$$

$$P_2 = a_{12} a_{23} a_{35}$$

$$\Delta = 1 - (a_{23} a_{32} + a_{23} a_{34} a_{42} + a_{23} a_{34} a_{45} a_{52} + a_{23} a_{35} a_{52} + a_{44})$$

$$+ (a_{23} a_{32} a_{44} + a_{23} a_{35} a_{52} a_{44})$$

$$\Delta_1 = 1 - 0$$

$$\Delta_2 = 1 - a_{44}$$

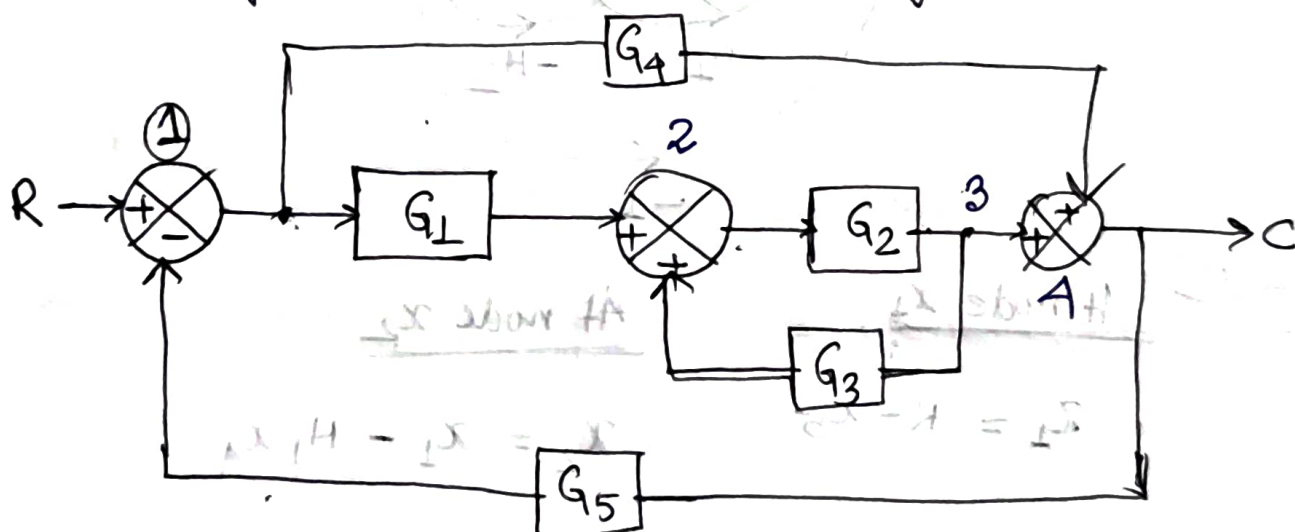
$$\therefore T = \frac{x_5}{x_1} = \frac{P_1 \Delta_1 + P_2 \Delta_2}{\Delta}$$

* Construction of Signal flow graph from Block diagram

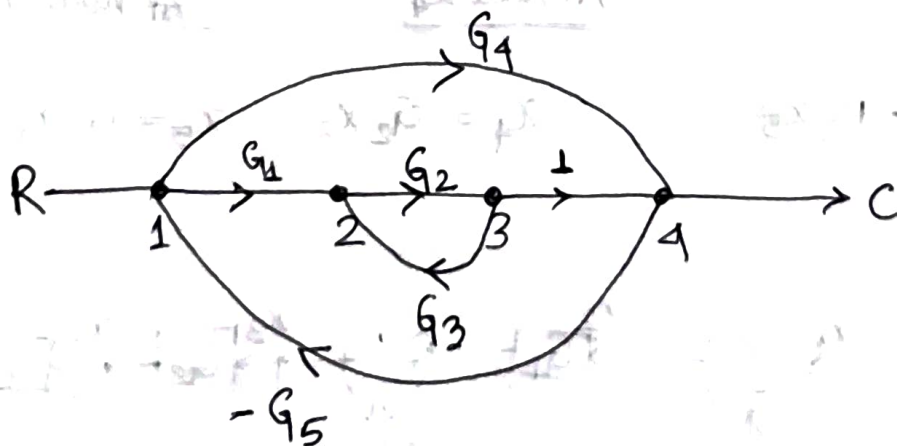
All variables ——— Summing points and takeoff points are represented by nodes

- ② If a summing point ^{is} displaced before a takeoff point in the direction of signal flow. In such case represent the summing point and take off point by a single ~~loop~~ node.
- ③ If a summing point is placed after a takeoff point in the direction of signal flow. In such case represent summing point and takeoff point by separate nodes connected by branch having transmittance signal unity.

e.g.:-



soln:



* Time Response of first order system:

$$a_0 \frac{dy}{dt} + a_1 y = b_0 r$$

$y \Rightarrow \text{o/p}$

$r \Rightarrow \text{i/p}$

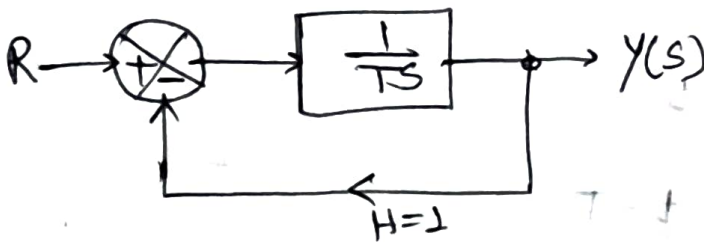
$$\Rightarrow \frac{a_0}{a_1} \frac{dy}{dt} + y = \frac{b_0}{a_1} r$$

$$\Rightarrow T \frac{dy}{dt} + y = K r$$

$$\Rightarrow ST Y(s) + Y(s) = K R(s)$$

$$\Rightarrow \frac{Y(s)}{R(s)} = \frac{K}{1 + ST}$$

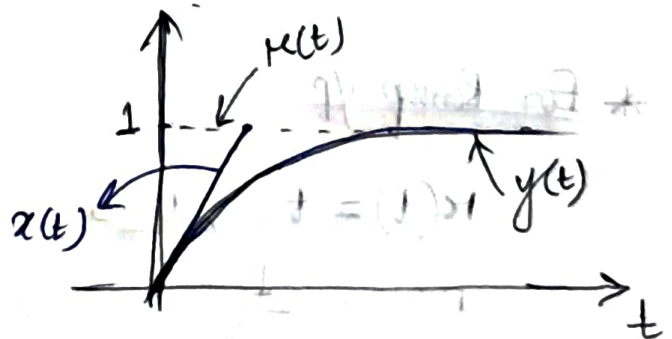
$$\left| \begin{array}{c} \frac{Ks}{1 + Ks} \\ \frac{1}{1 + TS} \end{array} \right|$$



$$\frac{Y(s)}{R(s)} = \frac{1}{1 + TS}$$

$\Rightarrow$ For unit step i/p.

$$r(t) = \begin{cases} 1 & ; t \geq 0 \\ 0 & ; t < 0 \end{cases}$$



$$R(s) = \frac{1}{s}, \quad s \neq 0$$

$$\therefore Y(s) = \frac{R(s)}{1 + TS} = \frac{1}{s(1 + TS)} = \frac{1}{s} - \frac{T}{1 + TS}$$

$$\therefore y(t) = (1 - e^{-t/T}) ; t \geq 0$$

$$\text{Error, } e(t) = r(t) - y(t)$$

$$= 1 - (1 - e^{-t/\tau})$$

$$= e^{-t/\tau}$$

$$\text{Steady state Error} = \lim_{t \rightarrow \infty} e(t) = \lim_{t \rightarrow \infty} e^{-t/\tau} = 0$$

$$\bullet \quad y(t) = (1 - e^{-t/\tau})$$

$$\left. \frac{dy}{dt} \right|_{t=0} = \left. \frac{e^{-t/\tau}}{\tau} \right|_{t=0} = \frac{1}{\tau}$$

Eqn of tangent :-

$$\cancel{y = \frac{1}{\tau} t} \quad x = \frac{1}{\tau} t$$

$$\text{When } x=1, \quad t=\tau$$

Time constant is the time reqd to reach the steady state value if the initial rate is maintained.

* For Ramp i/p

$$r(t) = t, \quad t \geq 0$$

$$R(s) = \frac{1}{s^2}$$

$$A = \lim_{s \rightarrow 0} \frac{d}{ds} \frac{1}{1+Ts}$$

$$= \lim_{s \rightarrow 0} \frac{-T}{(1+Ts)^2}$$

$$= -T$$

$$B = \frac{1}{1+Ts} \Big|_{s \rightarrow \infty} = 1$$

$$C = \frac{1}{s} \Big|_{s = -1/\tau} = -T$$

$$Y(s) = \frac{R(s)}{1+Ts} = \frac{1}{s^2(1+Ts)} = \frac{A}{s^2} + \frac{B}{s} + \frac{C}{1+Ts}$$

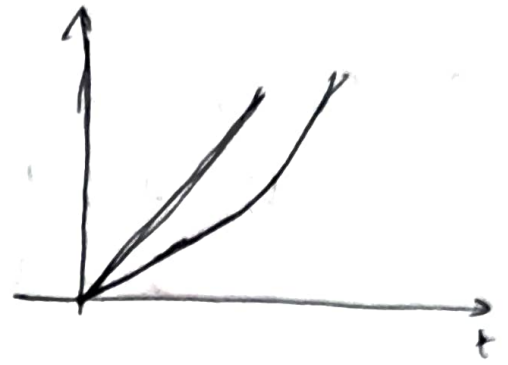
$$= \frac{1}{s} - T \left(\frac{1}{s^2} + \frac{1}{1+Ts} \right)$$

$$y(t) = t - T(1 - e^{-t/\tau}); \quad t \geq 0$$

$$\text{Error, } e(t) = r(t) - y(t)$$

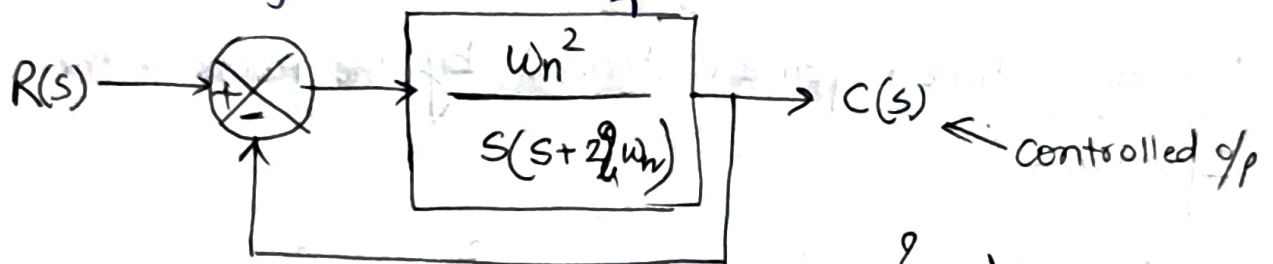
$$= t - (t - T(1 - e^{-t/T}))$$

$$= T(1 - e^{-t/T})$$



$$\therefore \text{Steady state error, } = \lim_{t \rightarrow \infty} T(1 - e^{-t/T}) = T$$

* Time Response of 2nd order system



ζ : damping ratio.

$$\text{Closed loop TF} = \frac{C(s)}{R(s)} = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)} \quad \omega_n: \text{Undamped natural freq. of oscillation.}$$

$$1 + \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}$$

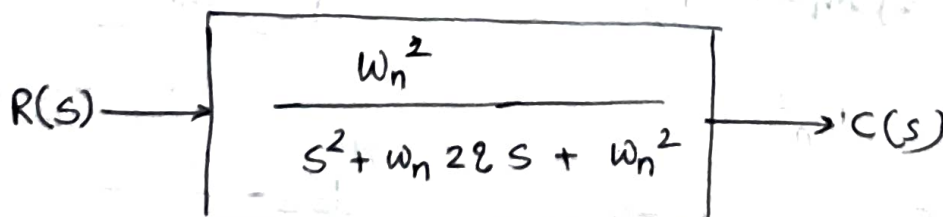
$$= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Forward path TF = G

Feedback TF = H

Loop TF = GH

CLTF = $\frac{G}{1+GH}$



$$C(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} R(s)$$

* Unit step response:-

$$r(t) = 1, \quad t \geq 0$$

$$0, \quad t < 0$$

$$R(s) = \frac{1}{s}$$

$$\therefore C(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} R(s)$$

Nature of time response is decided by the roots of the characteristic Equation-

$$\text{Char. Eqn} \rightarrow s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\therefore s = \frac{-2\zeta\omega_n \pm \sqrt{(2\zeta\omega_n)^2 - 4\omega_n^2}}{2}$$

$$s_{1,2} = -\cancel{2\zeta\omega_n} - \zeta\omega_n \pm j\omega_n \sqrt{1-\zeta^2}, \quad \boxed{\zeta < 1}$$

$$\therefore C(s) = \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)} + \frac{1}{s}$$

$$= \frac{\omega_n^2}{s(s-s_1)(s-s_2)}$$

$$C(s) = \frac{A}{s} + \frac{B}{s-s_1} + \frac{C}{s-s_2}$$

$$A = \frac{\omega_n^2}{(s-s_1)(s-s_2)} \Big|_{s=0} = \frac{\omega_n^2}{s_1 s_2}$$

$$B = \frac{\omega_n^2}{s(s-s_2)} \Big|_{s=s_1} = \frac{\omega_n^2}{s_1(s_1-s_2)}$$

$$C = \frac{\omega_n^2}{s(s-s_1)} \Big|_{s=s_2} = \frac{\omega_n^2}{s_2(s_2-s_1)}$$

Taking inverse Laplace T.

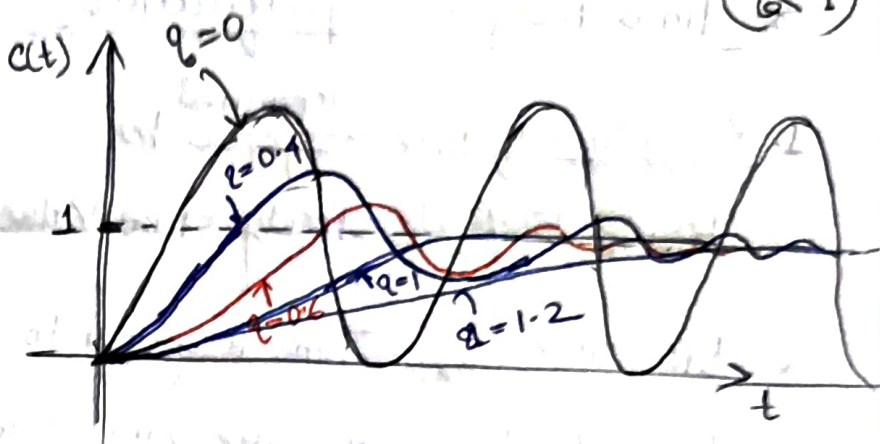
$$C(t) = A + B e^{s_1 t} + C e^{s_2 t}$$

After solving.

$$C(t) = 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin \left(\omega_n \sqrt{1-\zeta^2} t + \cos^{-1} \zeta \right); \quad t \geq 0 \quad (\zeta < 1)$$

When, $\zeta = 0$

$$C(t) = 1 - \cos \omega_n t$$



$$s_{1,2} = -\zeta \omega_n \pm j \omega_n \sqrt{1-\zeta^2}$$

$$= -\zeta \omega_n \pm j \omega_d$$

where

$$\omega_d = \omega_n \sqrt{1-\zeta^2}$$

damped freqⁿ osc.

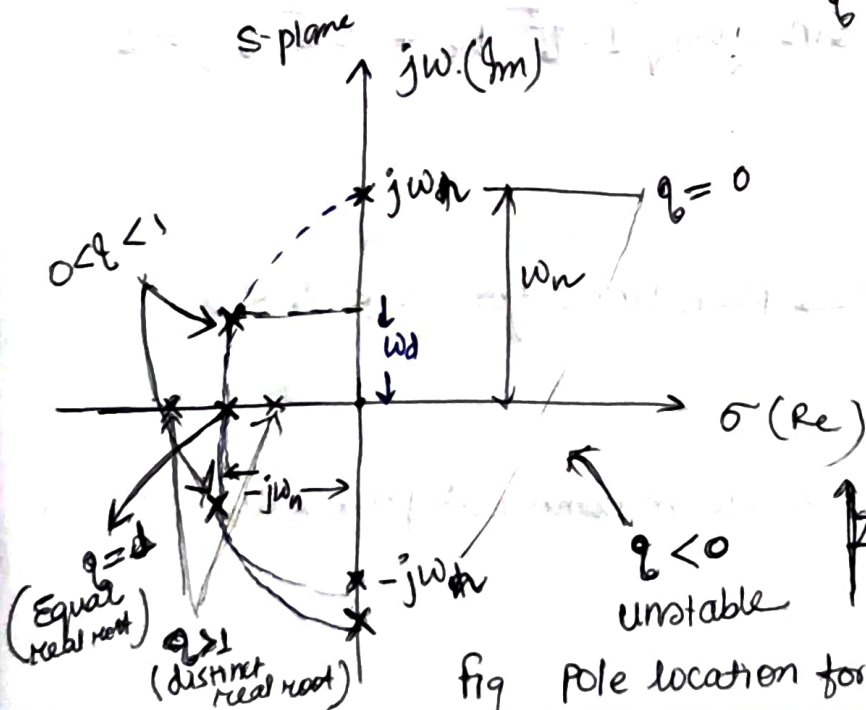
$\zeta = 0 \rightarrow$ Undamped response.
(Purely oscillating)

$0 < \zeta < 1 \rightarrow$ Underdamped "
(damped sinusoid)

$\zeta = 1 \rightarrow$ Critically damped "

$\zeta > 1 \rightarrow$ Overdamped "

$\zeta < 0 \rightarrow$ ~~overdamped~~ unstable/
negatively damped



When roots are on the
jw-axis $\rightarrow$ no damping
 $\rightarrow$ Response is purely sinusoidal

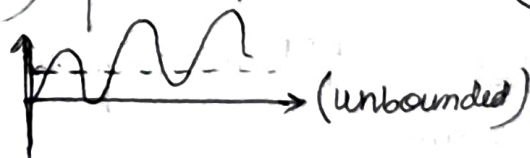
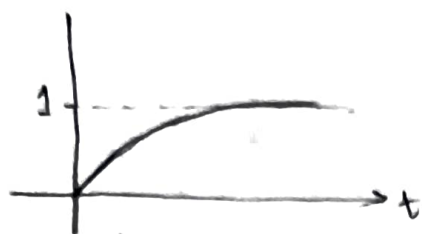


fig Pole location for a second order eqn for different ζ

eg:



1st order
($1 - e^{-t}$)



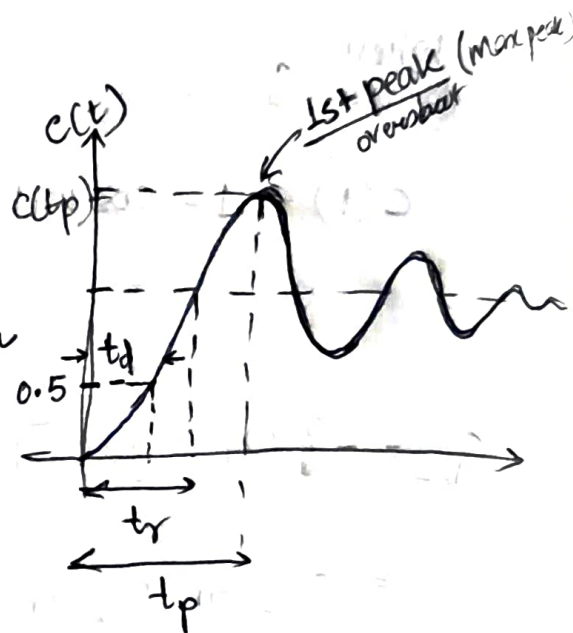
2nd order
critically or overdamped

* Time response Specification :-

1. Delay time (t_d) Time required to reach 50% of steady state value.

2. Rise time (t_r) :- Time required to reach 10-90% of steady state value for overdamped sys.

0-100% of the steady state value for underdamped sys.



$$C(t) = 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1 - \zeta^2}} \sin \left(\omega_n \sqrt{1 - \zeta^2} t + \cos^{-1} \zeta \right) = 1$$

$$t_r = \frac{\pi - \cos^{-1} \zeta}{\omega_n \sqrt{1 - \zeta^2}}$$

→ Rise time for $0 < \zeta < 1$

3. Peak time (t_p) Time reqd to reach maximum of peak value is called peak time.

$$C(t) = 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_n \sqrt{1-\zeta^2} t + \cos^{-1} \zeta)$$

$$\frac{dC(t)}{dt} = 0$$

After solving, General solution

$$t_p = \frac{n\pi}{\omega_n \sqrt{1-\zeta^2}} ; n=0, 1, 2, \dots$$

When, $n=1$ (1st overshoot) , When $n=2$ $\rightarrow$ (1st undershoot)

$$t_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

$$t_2 = \frac{2\pi}{\omega_n \sqrt{1-\zeta^2}}$$

4. Max peak overshoot (M_p):-

$$M_p = C(t_p) - 1$$

$$= \frac{e^{-\pi\zeta/\sqrt{1-\zeta^2}}}{\sqrt{1-\zeta^2}} e^{-\pi\zeta/\sqrt{1-\zeta^2}}$$

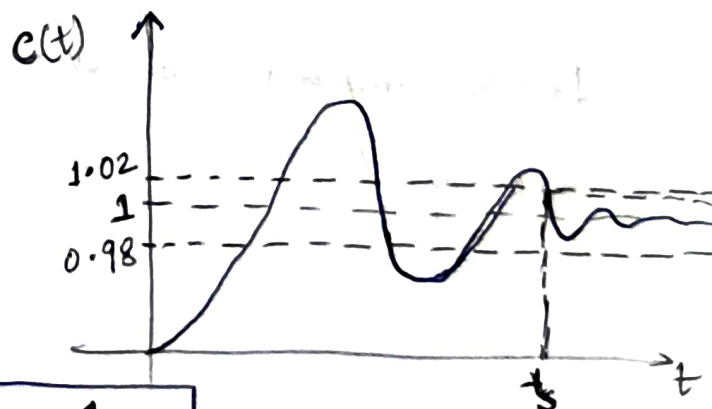
$$\% \text{ Maximum peak overshoot} = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\%$$

5. Settling time (t_s):-

$$1 \pm \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} = 1 \pm 2\%$$

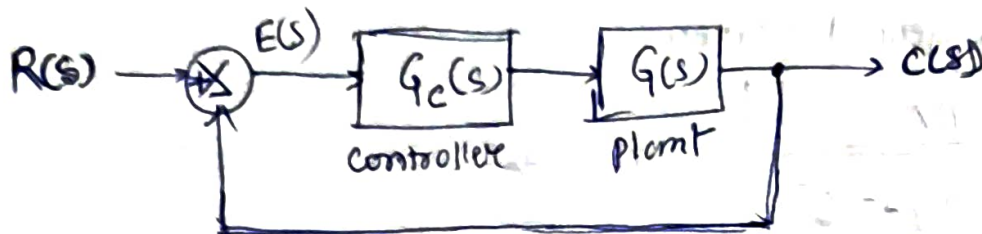
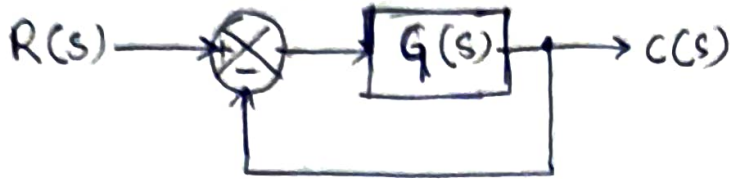
$$\frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} = 0.02$$

$$\Rightarrow e^{-\zeta \omega_n t} = 0.02 \Rightarrow t_s = \frac{4}{\zeta \omega_n}$$

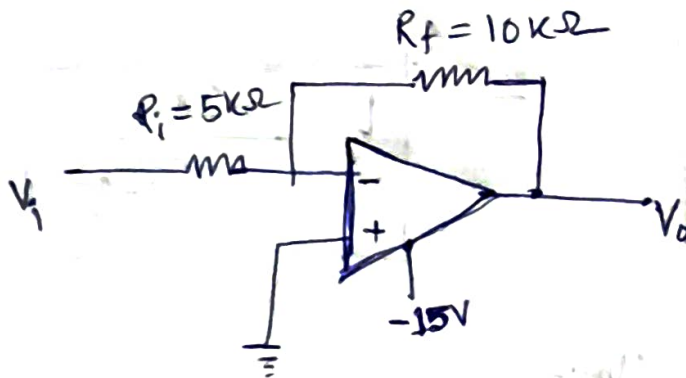


* Steady state error:-

$$E(s) = R(s) - C(s)$$



Assume that G_c uses an amplifier,

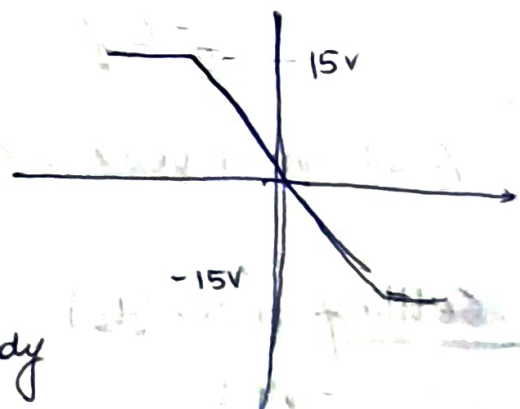


$$V_o = -\frac{R_f}{R_i} V_i$$

$$= -\frac{10}{5} V_i$$

$$= -2V_i$$

V_i	V_o
1	-2
2	-4
3	-6
4	-8
5	-10
10	-15



Non linearity of the controller causes steady state error

controlled q/p, $C(s) = \frac{G(s)}{1+G(s)} R(s)$

$$C(s) = E(s) G(s)$$

$$\therefore G(s) E(s) = \frac{G(s)}{1+G(s)} R(s)$$

$$E(s) = \frac{R(s)}{1+G(s)}$$

* Steady state error:-

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s E(s), \text{ where } E(s) = \mathcal{L}\{e(t)\}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \frac{R(s)}{1+G(s)}$$

, for linear system, steady state error depends on loop gain

* Unit step input:-

$$r(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$$

$$R(s) = \frac{1}{s}$$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} s E(s)$$

$$= \lim_{s \rightarrow 0} s \frac{R(s)}{1+G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{1+G(s)} = \frac{1}{1+G(0)}$$

$$= \frac{1}{1+k_p}$$

where

$k_p = \lim_{s \rightarrow 0} G(s) \rightarrow$ position error const.

* Ramp i/p

$$r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$R(s) = \frac{1}{s^2}$$

$$e_{ss} = \lim_{s \rightarrow 0} s R(s) \frac{1}{1+G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s + s G(s)}$$

$$= \frac{1}{\lim_{s \rightarrow 0} s G(s)} = \frac{1}{K_v}$$

where, $K_v = \lim_{s \rightarrow 0} s G(s)$

Velocity error const.

* parabolic i/p

$$r(t) = \begin{cases} \frac{t^2}{2}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$R(s) = \frac{1}{2s^3}$$

$$e_{ss} = \lim_{s \rightarrow 0} s R(s) \frac{1}{1+G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2 G(s)}$$

$$= \frac{1}{\lim_{s \rightarrow 0} s^2 G(s)}$$

$$= \frac{1}{K_a}$$

where

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)$$

= Acceleration error const.

* Consider the -2TF

$$G(s) = \frac{K (T_1 s + 1) (T_2 s + 1) \dots}{s^p (T_{p1} s + 1) (T_{p2} s + 1) \dots} \quad \text{(Time-constant form)}$$

type no. $\rightarrow$ No. of poles at origin

$$= p$$

$$= \frac{K' (s + T_1) (s + T_2) \dots}{s^p (s + P_1) (s + P_2) \dots} \quad \text{(pole-zero form)}$$

where $K' = K \frac{\pi_i T_{zi}}{\pi_j T_{pj}}$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G(s)}, \text{ steady state error is dominated by the no. of poles at origin.}$$

Type-0 sys:- , $n=0$

Unit step

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$U(s) = \frac{1}{s}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{1 + G(s)}$$

$$= \frac{1}{1 + K}$$

Ramp i/p

$$R(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$R(s) = \frac{1}{s^2}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \cdot \frac{1}{s^2}}{1 + G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s + s G(s)}$$

$$= \frac{1}{\lim_{s \rightarrow 0} s G(s)} = \frac{1}{K_v} = \infty$$

* Parabolic i/p

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \times \frac{1}{s^3}}{1 + G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2 G(s)}$$

$$= \infty$$

Type-n

$$G(s) = \frac{K (T_{z1}s+1) (T_{z2}s+1) \dots}{s^n (T_{p1}s+1) (T_{p2}s+1) \dots}$$

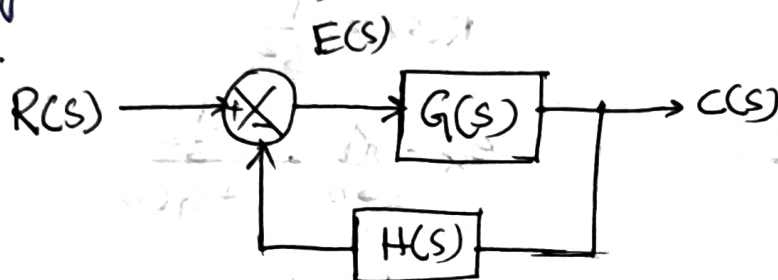
	Type-0	Type-1	Type-2
Step i/p	$\frac{1}{1+K}$	0	0
Ramp i/p	∞	$\frac{1}{K}$	0
Parabolic i/p	∞	∞	$\frac{1}{K}$

→ As the ~~steady~~ type no. (n) increases, steady state error reduces.

→ If we go beyond 2 (typeno.), steady state error will reduce by dynamic error increases and difficult to stabilize.

→ Also we will not get information, how the steady state error varies with time.

* Steady state error for non-unity feedback control system:-



$$C(s) = \frac{G(s)}{1 + G(s)H(s)} R(s) \quad \text{--- (1)}$$

$$E(s) = R(s) - C(s)H(s)$$

$$\therefore C(s) = E(s)G(s)$$

$$\text{(1)} \Rightarrow E(s) = \frac{R(s)}{1 + G(s)H(s)}$$

Steady state error,

$$e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \frac{R(s)}{1 + G(s)H(s)}$$

Q. For a unit feedback control system, the forward path transfer function is given by —

$$G(s) = \frac{20}{s(s+2)(s^2+2s+20)}, \text{ Determine the}$$

steady state error of the system, when the i/p are

- ① 5 ② 5t ③ $\frac{3t^2}{2}$

Soln:

$$e_{ss} = \lim_{s \rightarrow 0} \frac{sR(s)}{1+G(s)}$$

① $R(s) = \frac{5}{s}$

② $R(s) = \frac{5}{s^2}$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} \frac{s \times \frac{5}{s}}{1 + \frac{20}{s(s+2)(s^2+2s+20)}}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{\frac{5}{s}}{1 + \frac{20}{s(s+2)(s^2+2s+20)}}$$

$$= 0$$

$$= \lim_{s \rightarrow 0} \frac{5}{s + \frac{20}{(s+2)(s^2+2s+20)}}$$

③ $R(s) = \frac{3}{2s^3}$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \times \frac{3}{2s^3}}{1 + \frac{20}{s(s+2)(s^2+2s+20)}}$$

$$= \frac{\frac{3}{2s^2}}{\frac{20}{s(s+2)(s^2+2s+20)}}$$

$$= \lim_{s \rightarrow 0} \frac{\frac{3}{2s^2}}{1 + \frac{20}{s(s+2)(s^2+2s+20)}}$$

$$= \lim_{s \rightarrow 0} \frac{\frac{3}{2}}{\frac{s^2 + \frac{20s}{(s+2)(s^2+2s+20)}}} = \infty$$

Q. A 2nd order sys. has transfer function given by - (GATE 1991)

$$G(s) = \frac{25}{s^2 + 8s + 25} \text{ , if the sys. initially at rest}$$

is subjected to a unit step input at $t=0$, the 2nd peak in the response will occur at ~~5 sec~~ (a) π sec (b) $\pi/2$ sec (c) $2\pi/3$ sec (d) $\pi/3$ sec

Soln:- Stand. 2nd order sys. T.F -

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\therefore \omega_n = 5 \text{ Rad/s}$$

$$2\zeta\omega_n = 8$$

$$\Rightarrow \zeta = \frac{8}{2 \times 5} = 0.8$$

$$\text{Now, } t_p = \frac{n\pi}{\omega_d} = \frac{n\pi}{\omega_n \sqrt{1-\zeta^2}}$$

For 2nd peak, $n=3$

$$t_p = \frac{3\pi}{5 \sqrt{1-(0.8)^2}} \text{ sec} = \pi \text{ sec}$$

Q. The open loop T.F of a unity feedback sys. is given by -

$$G(s) = \frac{K}{s(1+st)} \text{ , where, } K \text{ and } T \text{ are positive const. By}$$

what factor should the amplifier gain be reduced so that the peak overshoot of unit step response of the sys. is reduced from 75% to 25%.

Soln:

$$\text{Closed loop TF. } \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

$$= \frac{K}{s(1+sT)} = \frac{K}{s^2T + s + K} = \frac{K/T}{s^2 + \frac{1}{T}s + \frac{K}{T}}$$

$$\text{Here } \omega_n^2 = \frac{K}{T}$$

$$\Rightarrow \omega_n = \sqrt{\frac{K}{T}} \text{ rad/sec}$$

$$2\zeta\omega_n = \frac{1}{T}$$

$$\Rightarrow \zeta = \frac{1}{2T \times \sqrt{\frac{K}{T}}} = \frac{1}{2\sqrt{KT}}$$

$$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$\text{Now, } e^{-\frac{\pi\zeta_1}{\sqrt{1-\zeta_1^2}}} = 0.75$$

$$e^{-\frac{\pi\zeta_2}{\sqrt{1-\zeta_2^2}}} = 0.25$$

$$\Rightarrow \zeta_1 = 0.09118$$

$$\Rightarrow \zeta_2 = 0.4037$$

$$\therefore \frac{\zeta_1}{\zeta_2} = \frac{\frac{1}{2\sqrt{KT}}}{\frac{1}{2\sqrt{K_2T}}} = \sqrt{\frac{K_2}{K}} = \frac{0.09118}{0.4037}$$

$$\boxed{K_1 = 19.6 K_2}$$

Stability:-

An LTI system is stable if the following two notions of stability are satisfied -

- ① If i/p is bounded, output is bounded (BIBO)
- ② If there is no i/p, o/p tends towards zero
- asymmetric stability.

$$\frac{d^2 y}{dt^2} + 2 \zeta \omega_n \frac{dy}{dt} + \omega_n^2 y = x(t)$$

Unbounded i/p $\Rightarrow$ unbounded o/p.

Then nothing can be said about such sys.

depends on -
initial condⁿ
sys parameters
Small change in the
i/p large change in o/p.

Pole position:

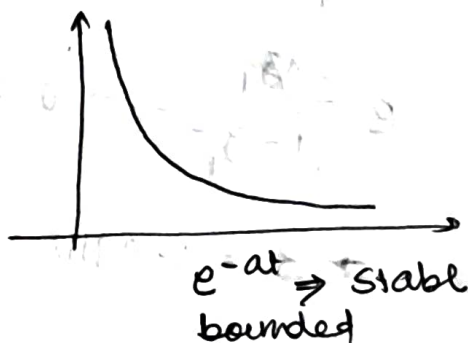


$$\frac{C(s)}{R(s)} = \frac{1}{s+a}$$

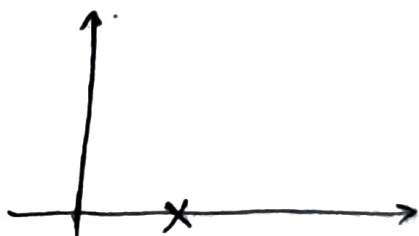
$$C(s) = \frac{1}{s+a}$$

$$\Rightarrow c(t) = e^{-at}$$

Impulse response

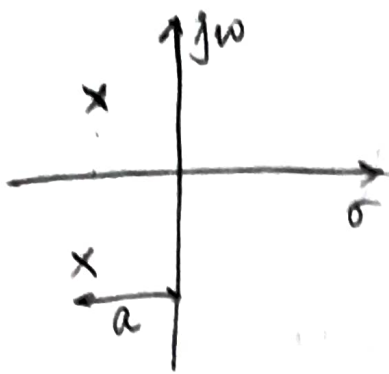


Pole position:



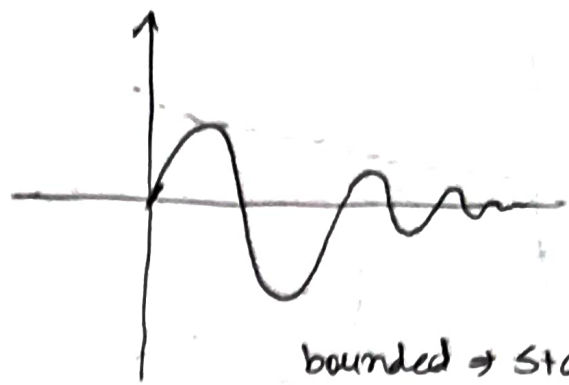
Impulse response





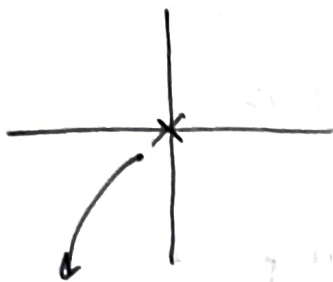
TF: $\frac{1}{(s+a)^2 + \omega_n^2}$

$C(t) = e^{-at} \sin \omega t$



bounded $\Rightarrow$ stable

$e^{-at} \sin \omega t$



TF = $\frac{1}{s}$

$C(s) = \frac{1}{s} R(s)$

$= \frac{1}{s} \cdot \frac{1}{s+1}$

$= \frac{1}{s} - \frac{1}{s+1}$

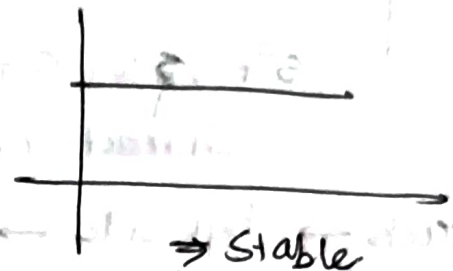
marginally stable;

root posⁿ same

for different $C(t) = (1 - e^{-t}) u(t)$

subjected signal

diff o/p; st or unit.

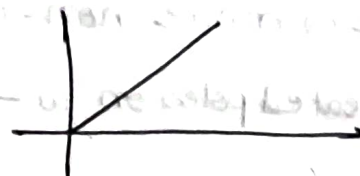


$\Rightarrow$ Stable

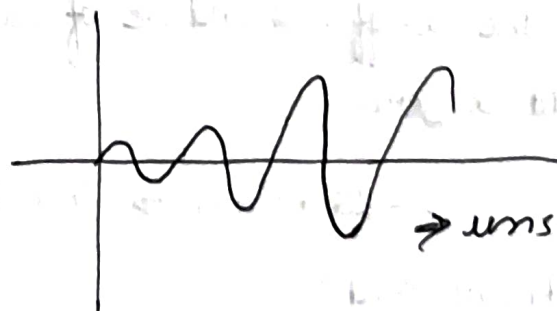
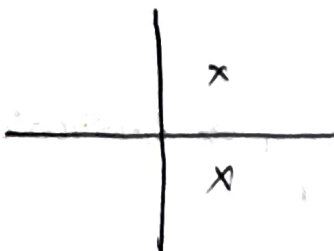
Now, subjected to unit step.

$C(s) = \frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2}$

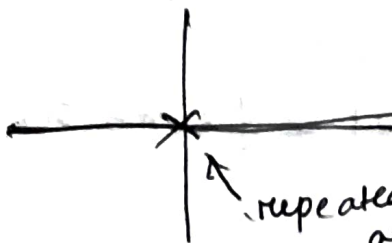
$C(t) = t u(t)$



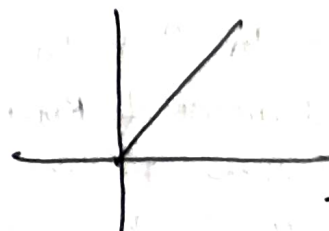
unstable.



$\Rightarrow$ unstable



repeated root at origin

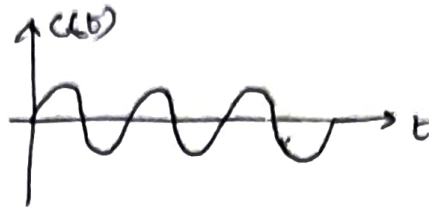


$\Rightarrow$ unstable

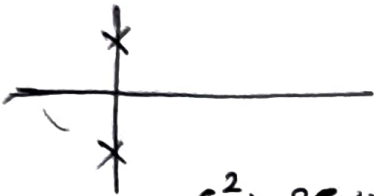


$$C(s) = \frac{1}{s^2 + \omega^2}$$

$$c(t) = \sin \omega t$$

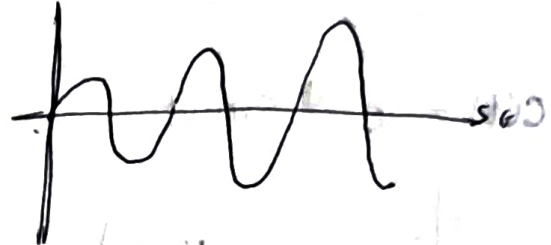


⇒ Stable
(marginally stable)



$$s^2 + 2\zeta \omega_n s + \omega_n^2 = 0$$

characteristic eqn.



⇒ Unstable

Roots → left side → stable

Roots → on $j\omega$ → single pair → marginally stable

Roots → on right → unstable.

Roots → on $j\omega$ (repeated) → unstable.

→ Conclusion: poles with negative real part stable system.

poles with positive real part unstable.

one or more non-repeated poles on $j\omega$ → marginally stable
or limited by stable

Repeated poles on $j\omega$ → unstable.

* Necessary conditions for stability:-

$$G(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_0 = 0$$

all the co-eff. should be of same sign (real) no. co eff. should be zero.

$$s^2 - 2s + 1 = 0 \Rightarrow \text{co-eff. different sign} \Rightarrow \text{unstable.}$$

* Sufficient condⁿ:-

$$G(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_0 = 0$$

first column elements of Routh array should be of some sign for the sys. will be stable.

if any sign changes → unstable

no. of sign changes indicated no. of roots in the right half of s-plane.

* Routh stability criterion:-

$$G(s) = a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_n = 0$$

s^n	a_0	a_2	a_4	...
s^{n-1}	a_1	a_3	a_5	...
s^{n-2}	b_1	b_3	b_5	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
s^0	$\vdots$	$\vdots$	$\vdots$	$\vdots$

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}$$

$$b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1}$$

first column
if same sign
→ stable.

$+a_0$
 $+a_1$
 $-b_1$
 $+b_2$
sign change = 2, means 2 roots in right side of s-plane.

Prob 1:- $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$

The Routh Table:

s^4	1	18	5
s^3	8	16	
s^2	$\frac{18-2}{1} = 16$	$\frac{5-0}{1} = 5$	
s^1	$27/16$	0	
s^0	5		

no sign change → stable

any row can be divided by common factor

Prob 2:- $3s^4 + 10s^3 + 5s^2 + 5s + 2 = 0$

s^4	3	5	2
s^3	10	5	0
s^2	$\frac{50-15}{10} = \frac{7}{2}$	2	
s^1	$-\frac{1}{7}$	0	
s^0	2	0	

$$\frac{7}{2} - 4 = -\frac{1}{2} \times \frac{x}{7}$$

Here is a sign change → unstable
There are two sign changes → two roots in Right side. s-plane.

Special Cases:

- ① When the 1st term in any row is zero, while rest of the row have atleast one non-zero term. (It has to first fulfill the necessary condition)

$$q(s) = s^n + () s^{n-p} + () \dots$$

$$\begin{array}{c|ccc} s^n & 1 & 2 & \dots \\ s^{n-p} & 1 & 2 & \dots \\ & 0 & \dots & \dots \end{array}$$

eg:- Consider the characteristic equation,

$$s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$$

The Routh array is -

$$\begin{array}{c|ccc} s^5 & 1 & 2 & 3 \\ s^4 & 1 & 2 & 5 \\ s^3 & 0(+) & -2 & \\ s^2 & \frac{2\epsilon+2}{\epsilon} & 5 & \\ s^1 & -\frac{4\epsilon-4-5\epsilon^2}{2\epsilon+2} & (-) & \\ s^0 & 5 & (+) & \end{array}$$

NOTE: If any one of the element is zero, then, s^2 elements will be undefined.

Because of this zero term, in the next row, the terms becomes infinite and Routh test breakdown.

Method: Substitute a small positive no. ' ϵ ' for the zero, and proceed to evaluate the rest of the Routh array, then examine the sign of the first column of Routh array by making ϵ tends to zero.

there is two sign changes $(+ \rightarrow - \rightarrow +)$

$\therefore$ The system is unstable, With two poles in the right-half s-plane: